PYTHON PROGRAMMING LANGUAGE

Python Became the Best Programming Language & fastest programming language. Python is used in Machine Learning, Data Science, Big Data, Web Development, Scripting, generative ai, prompt enginering, llm model, agentic ai, cloud also we are integration python. we will learn pyton from start to end || basic to expert. if you are not done programm then that is totally fine. I will explain from starting from scratch. python software - pycharm || vs code || jupyter || spyder

PYTHON INTERPRETTER

IDE (INTEGRATED DEVELOPMENT ENVIRONMENT)

PYTHON INTERPRETER --> What is Python interpreter? A python interpreter is a computer program that converts each high-level program statement into machine code. An interpreter translates the command that you write out into code that the computer can understand

PYTHON INTERPRETER EXAMPLE --> You write your Python code in a text file with a name like hello.py . How does that code Run? There is program installed on your computer named "python3" or "python", and its job is looking at and running your Python code. This type of program is called an "interpreter".

IDE (INTEGRATED DEVELOPMENT ENVIRONMENT) =>

- using IDE - one can write code, run the code, debug the code
- IDE takes care of interpreting the Python code, running python scripts, building executables, and debugging the applications.
- An IDE enables programmers to combine the different aspects of writing a computer program.
- if you wnated to be python developer only then you need to install (IDE -- PYCHARM)

PYTHON INTERPRETER & COMPILER

Both compilers and interpreters are used to convert a program written in a high-level language into machine code understood by computers. Interpreter -->

- Translates program one statement at a time
- Interpreter run every line item
- Execut the single, partial line of code
- Easy for programming

Compiler -->

- Scans the entire program and translates it as a whole into machine code.
- No execution if an error occurs
- you can not fix the bug (debug) line by line

Is Python an interpreter or compiler? Python is an interpreted language, which means the source code of a Python program is converted into bytecode that is then executed by the Python virtual machine. Python is different from major compiled languages, such as C and C + +, as Python code is not required to be built and linked like code for these languages.

ANACONDA

Anaconda is a distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment.

In [1]:
```
1 + 1 # ADDITION
```
Out[1]: 2

In [2]:
```
2-1
```
Out[2]: 1

In [3]:
```
3*4
```
Out[3]: 12

In [4]:
```
8 / 4 # Division
```
Out[4]: 2.0

In [5]:
```
8 / 5 #float division
```
Out[5]: 1.6

In [6]:
```
8/4 ## float division
```
Out[6]: 2.0

In [7]:
```
8 // 4 #integer division
```
Out[7]: 2

In [8]:
```
8 + 9 - 7
```
Out[8]: 10

In [9]:
```
8 + 8 - #syntax error:
```

```
  Cell In[9], line 1
    8 + 8 - #syntax error:
            ^
SyntaxError: invalid syntax
```

In [10]:
```
5 + 5 * 5
```

Out[10]: 30

In [11]:
```
(5 + 5) * 5 # BODMAS (Bracket || Oders || Divide || Multiply || Add || Substact)
```

Out[11]: 50

In [12]:
```
2 * 2 * 2 * 2 * 2 # exponentaion
```

Out[12]: 32

In [13]:
```
2 ** 5
```

Out[13]: 32

In [14]:
```
15 / 3
```

Out[14]: 5.0

In [15]:
```
10 // 3
```

Out[15]: 3

In [16]:
```
14 % 2 # Modulus
```

Out[16]: 0

In [17]:
```
15 %% 2
```

```
  Cell In[17], line 1
    15 %% 2
        ^
SyntaxError: invalid syntax
```

In [18]:
```
a,b,c,d,e = 15, 7.8, 'nit', 8+9j, True

print(a)
print(b)
print(c)
print(d)
print(e)
```

```
15
7.8
nit
(8+9j)
True
```

In [19]:
```
print(type(a))
print(type(b))
print(type(c))
print(type(d))
print(type(e))
```

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'complex'>
<class 'bool'>
```

In [20]:
```
type(c)
```

Out[20]:
```
str
```

- So far we code with numbers(integer)
- Lets work with string

In [21]:
```
'Redam Jaswanth'
```

Out[21]:
```
'Redam Jaswanth'
```

python inbuild function - print & you need to pass the parameter in print()

A function is a block of code which only runs when it is called. You can pass data, known as parameters, into a function. A function can return data as a result.

In [22]:
```
print('Redam Jaswanth')
```

```
Redam Jaswanth
```

In [23]:
```
s1 = 'Redam Jaswanth'
s1
```

Out[23]:
```
'Redam Jaswanth'
```

In [24]:
```
a = 2
b = 3

a + b
```

Out[24]:
```
5
```

In [25]:
```
c = a + b
c
```

```
Out[25]: 5
```

```
In [26]: a = 3
         b = 'Hi'
         type(b)
```

```
Out[26]: str
```

```
In [27]: a + b
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[27], line 1
----> 1 a + b

TypeError: unsupported operand type(s) for +: 'int' and 'str'
```

```
In [28]: print('Redam',  'Jaswanth')
```

```
Redam Jaswanth
```

```
In [29]: # print the nit 2 times
         'Jaswanth' + ' Jaswanth'
```

```
Out[29]: 'Jaswanth Jaswanth'
```

```
In [30]: #5 time print
         5 * 'Jaswanth'
```

```
Out[30]: 'JaswanthJaswanthJaswanthJaswanthJaswanth'
```

```
In [31]: 5 *' Jaswanth ' # soace between words
```

```
Out[31]: ' Jaswanth  Jaswanth  Jaswanth  Jaswanth  Jaswanth '
```

```
In [32]: print('c:\Jaswanth') #\n -- new line
```

```
c:\Jaswanth
```

```
In [ ]:
```